

Logistic Regression



Introduction

Logistic Regression is similar to linear regression, except it predicts whether something is **TRUE or FALSE**.

- ★ Instead of fitting a line to the data, logistic regression fits an "S" shaped (logit function).
- ★ This curve tells you the **probability** of this point belongs to the positive class.
- ★ Like linear regression, it can work with **multiple feature** (continuous and discrete).
- ★ Logistic regression models assume a **linear relationship between the independent variables and the log odds of the outcome**.

How to fit the data:

In Linear regression, we used **least square error** to find the line that **minimize the sum of the squares** of these residuals and use the residuals to calculate R^2 . while in Logistic regression we use **Maximum Likelihood**



Binary Classification

ماحول ال score الى بيفلح من ال Linear Regression
Probability من طريق ال Sigmoid Func

use Sigmoid Squashing function (Logistic Function)

$$\hat{y}_i = \frac{1}{1 + e^{-w^T X_i}} \quad \hat{y}_i \in [0, 1] \Rightarrow \text{Sigmoid produces prob}$$

Euclidean Loss (Squared Error)

$$\text{Loss} = \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

$$\text{Loss} = \sum_{i=1}^N \left(\frac{1}{1 + e^{-w^T X_i}} - y_i \right)^2$$



كثير من Minima
والتالي ممكن ال Training
بـ Converge عند Minima

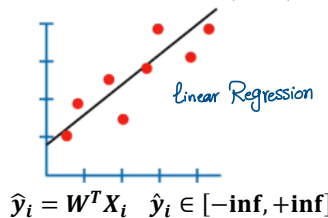
وتكون في نقطة ثانية فيها ال Cost أقل
Global Minima

- In order to Minimize the Loss **function use Gradient Descent (GD)** technique
- But this new Loss Function is **NOT** Convex function. GD Will not reach Global Minima, it may stuck on Local Minima

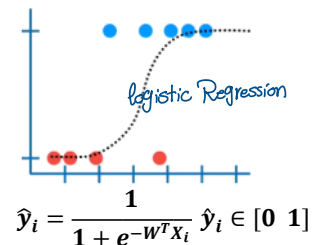
Since $\hat{y}_i \in [0, 1]$ We can consider $\hat{y}_i$ as Probability

- ☺ The higher the value of $\hat{y}_i$ the higher the probability that x_i belongs to the positive class (Class label = 1)
- ☺ The Probability of a given vector X_i Belongs to positive Class (y_1) is $\hat{y}_i$
- ☺ The Probability of a given vector x_i Belongs to negative class (y_0) is $(1 - \hat{y}_i)$

خلاص أنا كدة حسب ال $\hat{y}_i$
أزاي أحسب ال Coefficients
ال Line الى حسب ال Data Fit



$$\hat{y}_i = w^T X_i \quad \hat{y}_i \in [-\text{inf}, +\text{inf}]$$



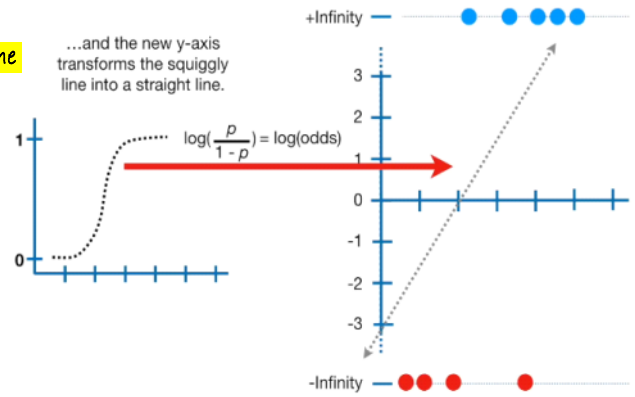
$$\hat{y}_i = \frac{1}{1 + e^{-w^T X_i}} \quad \hat{y}_i \in [0, 1]$$



calculate features coefficients

Coefficients in logistic regression represent the average **change in the log odds of the response** variable associated with a one-unit increase in the predictor variable.

...and the new y-axis transforms the squiggly line into a straight line.



في ال Linear Regression كان ال slope line هو ال Coef ال Feature

وهو عبارة عن التغير في ال y ← Change in response

ولكن في ال Logistic Reg هو عبارة عن Change in log odd of response

odds the ratio of the probability of the event happening to the probability of it not happening

Example If the probability of a customer buying a product is 0.7, then the odds of buying the product are $\frac{0.7}{(1-0.7)} = 2.33$.

Features with Continuous values

Just transform the y-axis in logistic regression is transformed from the "probability of y" to the "log (odds of y)" scale.

1 Initialize weights (Coef) (w)

For points:

2 Calculate $\hat{y} \rightarrow \frac{1}{1 + e^{-wt}} = p$

3 Calculate log odd $\rightarrow \log\left(\frac{p}{1-p}\right)$

4 Calculate Cost of the Current model (Fitted Line)

Plot line with all log odd values

Slope = Coeff of this Feature

intercept = w_0 (bias)

5 Repeat From 2 $\rightarrow$ 4 to min Cost

Example with Continuous values

أزاي هاحسب ال Coefficient

use a continuous variable (like weight) to predict obesity.

* **Estimated Intercept:** When weight = 0, the log (odds of obesity) are -3.48. The standard error for this estimate is 2.364.

* **Slope (Weight):** For every one unit of weight gained, the log (odds of obesity) increase by approximately 1.83. The standard error for this slope is 1.088.

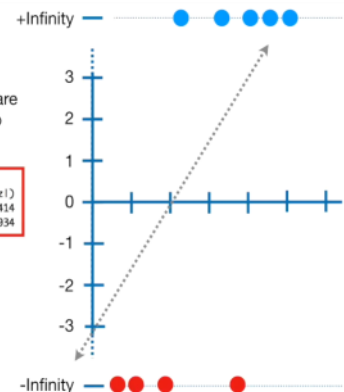
* **Z-value for Intercept:** It's -1.471, indicating that the estimated intercept is not statistically significant (since it's less than 2std away from zero, p-value < 0.05).

* **Z-value for Slope:** It's 1.678, also not statistically significant (since it's less than 2std away from zero, p-value < 0.05).

$y = -3.48 + 1.83 \text{ weight}$

The coefficients for the line are what you get when you do logistic regression.

Coefficients:				
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.476	2.364	-1.471	0.1414
weight	1.825	1.088	1.678	0.0934



الفهم أرجع ال hypothesis testing

Features with Discrete values

Use discrete variable like "whether or not an object has a mutated gene" to predict obesity.

- This type of logistic regression is very similar to how a **t-test** is done using linear models.

1. Transform the y-axis from the probability of being obese to the log (odds of obesity).

2. Calculate the log (odds of obesity) for an object with the normal gene --> log (odds gene normal) --> **Orange line**

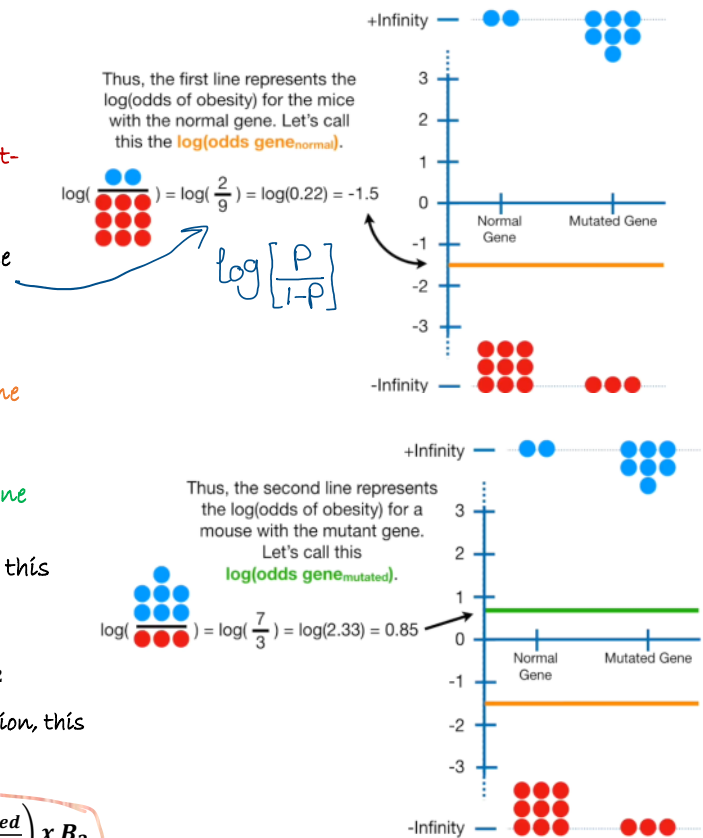
3. Calculate the log (odds of obesity) for an object with the mutant gene --> log (odds gene mutated) --> **Green Line**

4. These two lines come together to form the coefficients in this equation:

$$\text{Size} = \log(\text{odds gene}_{\text{normal}}) \times B_1 + \log(\text{odds gene}_{\text{mutated}} - \text{odds gene}_{\text{normal}}) \times B_2$$

5. subtracting one log from another can be converted into division, this term is log (odds ratio).

$$\text{Size} = \log(\text{odds gene}_{\text{normal}}) \times B_1 + \log\left(\frac{\text{odds gene}_{\text{mutated}}}{\text{odds gene}_{\text{normal}}}\right) \times B_2$$



Example with Continuous values

* **Estimated Intercept:** The value of -1.50 represents the log (odds) for individuals with the normal gene.

$$\text{size} = -1.5 \times B_1 + 2.35 \times B_2$$

* **GeneMutant Term:** The value of 2.35 is the log (odds ratio), indicating how much having the mutated gene affects the odds of being obese.

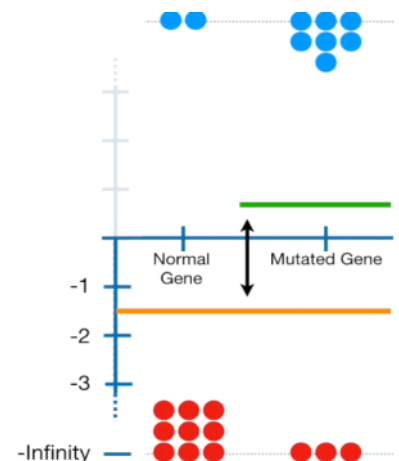
* **Standard Errors:** The standard error for the estimated intercept is 0.7817, and for the geneMutant, it's 1.0427.

Coefficients:				
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.5041	0.7817	-1.924	0.0544
geneMutant	2.3514	1.0427	2.255	0.0241

* **Z-value for Intercept:** it is -1.924 suggests it's not significantly different from 0 (confirmed by a p-value > 0.05).

* **Z-value for Slope:** for geneMutant (2.255) indicates statistical significance (p-value < 0.05).

DOUBLE BAM!!!



عرفت هأكمل Prediction في طريق وضع ال score في Sigmoid
عرفت أراي بنحسب ال Coef ال Features في طريق ال log odd
عرفت آت بعل update coef في طريق ال GD في الأتابة ال بيقال ال Cost
هنغير ال جلة دي شان قول ال Convex ال Func maximum Likelihood



calculate Objective (Loss function, Likelihood)

$$\log\left(\frac{p}{1-p}\right) = \log(\text{odds})$$

Exponentiate both sides:

$$\frac{p}{1-p} = e^{\log(\text{odds})}$$

Multiply both sides by $(1-p)$:

$$p = (1-p)e^{\log(\text{odds})}$$

$$p = e^{\log(\text{odds})} - pe^{\log(\text{odds})}$$

$$p + pe^{\log(\text{odds})} = e^{\log(\text{odds})}$$

$$p(1 + e^{\log(\text{odds})}) = e^{\log(\text{odds})}$$

Divide both sides by $(1 + e^{\log(\text{odds})})$

$$p = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$

The two cases can be combined in one equation as:

$$P(y_i | X_i) = \hat{y}_i^{y_i} * (1 - \hat{y}_i)^{1-y_i}$$

ما احتمال انتماء النقطة الى الكلاس y_i

$$P(y_i | X_i) = \hat{y}_i^{y_i} + (1 - \hat{y}_i) * (1 - y_i)$$

if $y_i = 1$ then $P(y_i | X_i) = \hat{y}_i$ if $y_i = 0$ then $P(y_i | X_i) = 1 - \hat{y}_i$

الصح ← Pred

For ALL N Samples (Assume All Samples are independent)

Loop over points

$$\rightarrow y_i = 0 \text{ and } \hat{y}_i = 0.3 \Rightarrow P = (0.3)^0 (1-0.3)^1 = 0.7$$

$$\rightarrow y_i = 1 \text{ and } \hat{y}_i = 0.7 \Rightarrow P = (0.7)^1 (1-0.7)^0 = 0.7$$

لنلاحظ الرقم 0.7 وهو كبير
في حالتين حالات
Correct Pred

$$\text{Likelihood} = \prod_{i=1}^N P(y_i | X_i) = \prod_{i=1}^N \hat{y}_i^{y_i} * (1 - \hat{y}_i)^{1-y_i}$$

من أجل كل point وحده حاصل العينة
يعمل Prob بشكل كبير

$$\rightarrow y_i = 0 \text{ and } \hat{y}_i = 0.7 \Rightarrow P = (0.7)^0 (1-0.7)^1 = 0.3 \Rightarrow \text{wrong Prediction as } \hat{y}_i > 0.5 = 1 \text{ while } y_i = 0$$

$$\rightarrow y_i = 1 \text{ and } \hat{y}_i = 0.3 \Rightarrow P = (0.3)^1 (1-0.3)^0 = 0.3 \Rightarrow \text{wrong Prediction as } \hat{y}_i < 0.5 = 0 \text{ while } y_i = 1$$

Objective is to Maximize the likelihood == Minimize the negative of the Likelihood (Loss Function)

Required to minimize $\rightarrow -\prod_{i=1}^N P(y_i | X_i) = -\prod_{i=1}^N \hat{y}_i^{y_i} * (1 - \hat{y}_i)^{1-y_i}$

هنا حاصل ضرب جميع الاحتمالات للنقاط وهنا الاحتمال منه قد ايه ان ينتمي الكلاس الى

وهذه الفانكشن هي اونها Convex

Instead we can minimize the "log" of the above Loss function

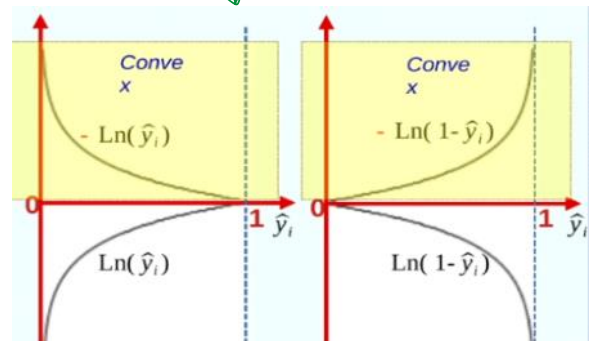
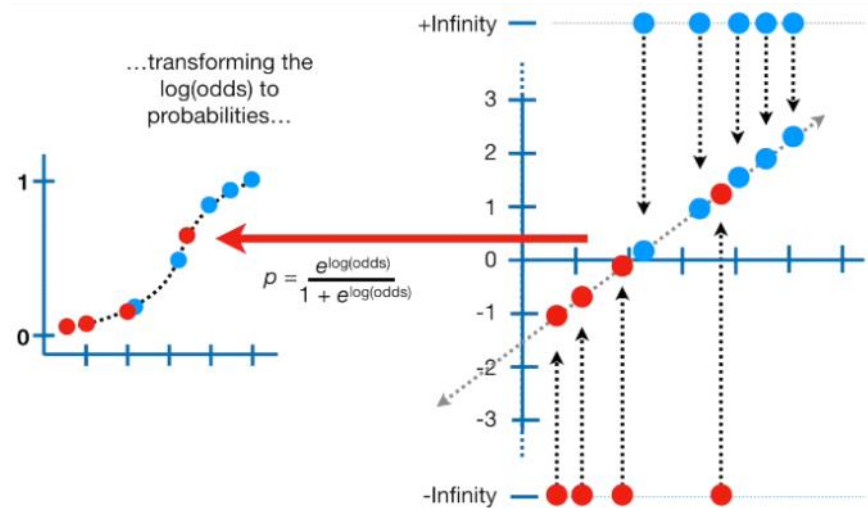
$$\text{Log Likelihood} = \ln\left(-\prod_{i=1}^N \hat{y}_i^{y_i} * (1 - \hat{y}_i)^{1-y_i}\right)$$

$$\text{Log Likelihood} = -\sum_{i=1}^N \ln(\hat{y}_i^{y_i} * (1 - \hat{y}_i)^{1-y_i})$$

$$\text{Log Likelihood} = -\sum_{i=1}^N \ln(\hat{y}_i^{y_i}) + \ln(1 - \hat{y}_i)^{1-y_i}$$

$$\text{Log Likelihood} = -\sum_{i=1}^N y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)$$

$$\text{Objective function } (J(w)) = -\frac{1}{N} \sum_{i=1}^N y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)$$



$$J(w) = -\frac{1}{N} \sum_{i=1}^N \text{Ln}(\hat{y}_i) \quad \text{when} \quad y_i = 1$$

$$J(w) = -\frac{1}{N} (1 - y_i) \text{Ln}(1 - \hat{y}_i) \quad \text{when} \quad y_i = 0$$



Gradient of Objective (Loss function, Likelihood)

$$\nabla J(w) = -\frac{1}{N} \sum_{i=1}^N y_i \frac{1}{\hat{y}_i} X_i \hat{y}_i (1 - \hat{y}_i) - (1 - y_i) \frac{1}{(1 - \hat{y}_i)} X_i \hat{y}_i (1 - \hat{y}_i)$$

$$\nabla J(w) = -\frac{1}{N} \sum_{i=1}^N y_i X_i (1 - \hat{y}_i) - (1 - y_i) X_i \hat{y}_i$$

$$\nabla J(w) = -\frac{1}{N} \sum_{i=1}^N y_i \frac{1}{\hat{y}_i} \frac{\partial \hat{y}_i}{\partial w} + (1 - y_i) \frac{1}{(1 - \hat{y}_i)} \frac{\partial (1 - \hat{y}_i)}{\partial w}$$

$$\hat{y}_i = \frac{1}{1 + e^{-w^T X_i}}$$

$$\frac{\partial \hat{y}_i}{\partial w} = \frac{X_i e^{-w^T X_i}}{(1 + e^{-w^T X_i})^2}$$

$$\frac{\partial \hat{y}_i}{\partial w} = X_i \left(\frac{1}{(1 + e^{-w^T X_i})} - \frac{1 + e^{-w^T X_i} - 1}{(1 + e^{-w^T X_i})^2} \right)$$

$$\frac{\partial \hat{y}_i}{\partial w} = X_i \hat{y}_i (1 - \hat{y}_i) \quad \text{similarity}$$

$$\frac{\partial (1 - \hat{y}_i)}{\partial w} = -X_i \hat{y}_i (1 - \hat{y}_i)$$

$$\nabla J(w) = -\frac{1}{N} \sum_{i=1}^N X_i (y_i - y_i \hat{y}_i - \hat{y}_i + y_i \hat{y}_i)$$

$$\nabla J(w) = -\frac{1}{N} \sum_{i=1}^N X_i (y_i - \hat{y}_i)$$

$$\nabla J(w) = \frac{1}{N} \sum_{i=1}^N X_i (\hat{y}_i - y_i)$$

$$\nabla J(w) = X_i (\hat{y}_i - y_i)$$

$$W^{\text{updated}} = W^{\text{old}} - \eta \nabla J$$